2.7 Step 5: Landslide risk assessment conclusion

2.7.1 Summary

This study has been undertaken to assess landslide risk in the sub-catchments of Rwandex-Magerwa, Bishenyi, Rwabayanga and Rusizi. The deliverable has been landslide risk maps for each watershed (refer to pdf maps accompanying report). Maps are classified in several classes.
The study has produced landslide risk maps in the form:

- Landslide inventories (step 1);
- Landslide susceptibility maps (step 2);
- Landslide hazard maps (step 3);
- Exposure databases (step 4)

2.7.2 Conclusion

The risk maps are here visual combinations of landslide process scenarios and land use categories. They allow to highlight the places where the risk could be the more problematic. Depending on the slope instability processes and the land use, the risk is different. For example, places where deep-seated landslides are located are areas where ground deformations are expected to be the larger (independently from soil creep, which is not discussed here, although highlighted in step 1). Such deformations can be very slow (a few centimetres per year or even less; e.g. Nobile et al., 2019; Dille et al., 2021) and therefore be not at all a problem for agricultural land. On the contrary, building a new road or heavy infrastructures on such areas could create problems (fractures, etc.) as most foundations cannot be deep enough to reach a stable bedrock below the surface of rupture of the landslides. While slow-moving deformations can have pervasive impacts difficult to mitigate, people are not in immediate dangers.

The hazard scenario we have computed here concerns shallow landslide processes. Their impacts will be much different from those of deep-seated processes. In terms of direct impacts, such shallow landslides can directly cover a road, but will not really damage it. Concerning big buildings with deep foundations, such landslides will also have minor impacts. However, shallow landslides can potentially remove a lot of soil and therefore have impacts of the fertility level of the agricultural land. Shallow landslides are in the region commonly associated with heavy rainfalls. They can occur very quickly and be responsible of fatalities.

When looking at such documents, we must be aware that further investigations are needed to really assess the problem, not only in terms of landslide process understanding (location, mechanism, deformation rate), but also in terms of vulnerability and direct and indirect impacts.